

In Situ Measurements of Giant Kelp (*M. pyrifera*) Motion

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Key Points:

- Currents induced significant tilts (up to 57°) over timescales longer than 5 min
- Surface gravity waves generated the strongest amplitude of tilting near the bottom
- Removal of the surface canopy amplified motion associated with both currents and waves

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Giant kelp (*Macrocystis pyrifera*), common in temperate environments, are buoyant, flexible, and surface-canopy forming. The three-dimensionally complex structure of giant kelp modifies local hydrodynamics by exerting a drag force on currents and dissipating wave energy. Their high flexibility and dynamic motion causes time-varying drag forces, but the underlying mechanisms remain poorly understood. Accurately parameterizing the time-varying drag of giant kelp in ocean models is essential for better quantifying the influence of kelp forests on coastal hydrodynamics. To quantify how giant kelp alters flow, we must first understand how they move in response in varying water levels, currents, and wave conditions. In this study, we used pressure sensors and 9-axis inertial measurement units to measure the motion of two giant kelp plants under combined forcing from tides, currents, and waves. Our findings reveal that strong currents can induce significant tilts on unmanipulated kelp—up to 57° from vertical positioning—over longer timescales (greater than 5 min). On shorter, surface gravity wave timescales (less than 30 s), the range of measured tilts was largest near the seafloor. Wave motion can also enhance the average tilt response of kelp to currents. Additionally, removing the surface canopy increases kelp motion near the surface across both shorter and longer timescales, although this effect attenuates with depth. Despite tidal depth variations of up to 2.6 m, which changes the extent of the kelp surface canopy, there was no clear relationship between water depth and kelp tilt.

Plain Language Summary In this study, we measured the motion of giant kelp plants to better understand how its unique structure and flexibility might affect ocean currents and wave propagation. We found that over long periods, strong currents can induce large tilts in kelp—up to 57°. Waves amplify motion near the seafloor and increase the amount of tilting induced by the currents. When the surface canopy was removed from one plant, it increased tilting with both currents and waves near the top of the plant. These findings will help us better understand how giant kelp interacts with ocean flows, which is important for predicting flow transport processes in the nearshore environment.

1. Introduction

Globally, kelp forests provide ecosystem services and economic value that generate up to \$562 billion U.S. dollars a year (Eger et al., 2023). In addition to supporting biodiversity and nearshore fisheries, they have the potential to remediate anthropogenic changes by sequestering carbon dioxide and buffering against ocean acidification (Dayton et al., 1984; Froehlich et al., 2019; Gao et al., 2021; Hoegh-Guldberg et al., 2023; Schiel & Foster, 2015). Kelp forests are also a culturally significant resource, and many communities around the world rely on kelp forests or farms as a source of food and fertilizer (Kain & Dawes, 1987; Thornton, 2015). More recently derived products such as alginate, livestock feed, and biofuels have increased industry interest in utilizing kelp as well (Frieder et al., 2022; Milledge et al., 2014; Peteiro, 2017).

Macrocystis pyrifera, or giant kelp, is a foundational species widely found along the coasts of west North America, South America, southern Australia and New Zealand, and southern Africa (Graham et al., 2007; Krumhansl et al., 2016; Wernberg et al., 2019). Giant kelp morphology consists of anywhere between 2 and 100 fronds attached to a single holdfast, which anchors the plant to the substrate (Jackson et al., 1985). Each frond is made up of a highly flexible stipe with many blades. At the base of each blade is a pneumatocyst, or gas bladder, which gives giant kelp its buoyant nature. Once fronds reach the surface, they form a dense canopy. Giant kelp is one of the largest and fastest-growing kelp species, often reaching lengths of 20 m or more (Abbott et al., 1992).

The surface canopy of giant kelp can modify the local physical and chemical environment near the surrounding near-surface environment. For example, the canopy absorbs heat from solar radiation and photosynthesizes at higher rates relative to the rest of the water column. This results in a surface layer with higher temperature,

dissolved oxygen, and pH, and lower carbon dioxide (Kowek et al., 2017; Traiger et al., 2022; Wu, Hannah, Petrie, et al., 2017). The dynamic motion of the surface canopy and changes in the extent of canopy coverage can create fluctuations in the amount of light that reaches understory algae at the seafloor (Wing et al., 1993). The surface canopy also attenuates short wave energy (Lindhart et al., 2024). Since surface canopy coverage is at maximum during late summer and significantly reduced in winter, the physical and ecological effects of the surface canopy vary with both depth and season. While giant kelp is a species of brown macroalgae, we follow the convention established in the literature (e.g., see Elsmore et al., 2024; Mullarney & Pilditch, 2017; Utter & Denny, 1996) and henceforth use the term “plant” to refer to giant kelp individuals (all the stipes and fronds attached to a single holdfast).

In the water column, giant kelp forests alters hydrodynamic conditions by modifying currents and wave energy. Giant kelp forests reduce currents and moderately dampen higher-frequency internal waves and surface gravity waves (Elsmore et al., 2022; Jackson & Winant, 1983; Lindhart et al., 2024; Monismith et al., 2022; Rosman et al., 2007). Changes in current magnitude and direction alters the transport of nutrients, spores, and larvae and the population connectivity between different kelp patches (Gaylord et al., 2012). While kelp forests attenuate average current speeds, they can also generate flow variability by causing preferential flow through channels between kelp individuals and suppressing headland upwelling (Valle-Levinson et al., 2022). Gaylord et al. (2007) also reported faster currents on the edge and outside of a kelp forest.

The degree to which giant kelp modifies the surrounding environment is governed by ambient hydrodynamic conditions. For example, Traiger et al. (2022) found that the greatest modifications to surface water chemistry were in locations with lower wave heights and slower currents. Tidal cycles also play a role, as the length of the surface canopy is longest at low tide and shortest at high tide. Lindhart et al. (2024) showed that because of this tidal change in canopy extent, attenuation coefficients for short waves were significantly higher during low tide intervals. The influence of kelp forests on the flow also strongly depends on the response of giant kelp motion to local hydrodynamics. It is the unique morphology of giant kelp—flexible stipes and buoyant pneumatocysts—that allows individuals to move with the surrounding flow. Generally, flexibility is an evolutionary strategy employed by seagrasses and macroalgae to reduce drag forces and increase survivability in high-flow environments (Utter & Denny, 1996; Vogel, 1984).

It is well known that the reduction in drag forces in comparison to a rigid plant allows giant kelp to minimize the energy dissipated from the water and the extent that kelp modifies the flow. In a dynamically scaled laboratory model, Rosman et al. (2013) found that wave-current interactions significantly increased the drag forces acting on kelp fronds, suggesting that kelp motion with surface gravity waves plays a non-negligible role in kelp forest transport and circulation. Mullarney and Pilditch (2017) measured the motion of giant kelp under wave forcing and observed that giant kelp had contrasting depth-dependent responses to waves of different periods: stronger tilting with surface gravity waves occurs at the bottom of the kelp (near the holdfast), while the response to infragravity waves is intensified near the surface. These results imply that the amount of energy dissipated by kelp, which is dependent on its detailed motion, may be both depth and frequency dependent. Thus, to develop accurate predictions for flows through kelp forests, it is necessary to first understand how kelp motion responds to varying depth, wave, and current conditions.

Theoretical and numerical models have been developed to link the motion of flexible plants to the amount of energy dissipated from the flow. Luhar and Nepf (2011) predicts the deflection of aquatic plants in steady currents by balancing hydrodynamic forcing with the restoring force from plant buoyancy and stiffness. As plants reconfigure from the flow, they experience reduced drag because their shape becomes more streamlined and a smaller frontal area is exposed to the flow (De Langre, 2008). This framework was extended to predict the drag reduction of stiff vegetation (e.g., blade stiffness is the dominant restoring force instead of buoyancy) in oscillatory flows, where more flexible motion correlates with decreased relative velocity between the plant and surrounding flow (Luhar & Nepf, 2016). Similarly, Henderson (2019) predicted depth-dependent stem motion and wave dissipation for highly flexible and buoyant vegetation like giant kelp and found that stem tilts are strongest near the seafloor. However, it is still unclear how cross-interactions between depth, currents, and waves can influence kelp motion and drag, and the influence of complex geometry like that of the surface canopy of giant kelp has also not been studied in detail. Modeling approaches to study flexible plants in oscillatory flows often do not account for the effects of inertial forces (resulting from the acceleration of the fluid under wave forcing), which Meng and Chamecki (2025) showed to be an important control for giant kelp motion in their natural

environments. Additionally, the large contribution of biomass and buoyancy from the surface canopy could be a strong control for the overall flexibility of giant kelp.

Given the uncertainties in kelp motion and how hydrodynamic conditions affect the drag of giant kelp, coastal circulation models often neglect the impact of kelp forests on bottom stress or use oversimplified parameterizations that are independent of tides, waves, and currents, limiting our understanding of how kelp forests alter coastal flow processes. One study on simulating the effects of kelp forests on tidal circulation used three simple constants for the bulk drag coefficient that were estimated from laboratory models instead of real kelp forests (Wu, Hannah, O'Flaherty-Sproul, & Thupaki, 2017). In a model of a giant kelp farm, Yan et al. (2021) utilized a drag coefficient derived from Utter and Denny (1996) that is for a single kelp frond with no surface canopy. Neither of these studies account for the effects of kelp motion on drag reduction. Strong-Wright and Taylor (2024) took a more detailed approach and applied a three-point numerical model for the motion of a single giant kelp frond, including the surface canopy, in their tidal circulation model. They directly calculate the drag force from the kelp, which is then implemented in their large eddy simulations. To date, only a few studies have attempted to explicitly incorporate surface canopy effects in an ocean model or address the more complex dynamics driven by the interactions between giant kelp and surface waves (Fagundes et al., 2023; Zhu et al., 2021).

In order to develop accurate parameterizations for the drag of giant kelp, we must first understand its dynamic motion in variable hydrodynamic conditions. In this paper, we present results from a before-after control-impact experiment to investigate how (a) kelp individuals move in different wave and current regimes, and (b) how the presence of the surface canopy affects kelp motion. Measurements of kelp motion were collected using 9-axis inertial measurement units (IMU) that measure raw acceleration, angular rate, and magnetic data. Combined with measurements of tidal depth, current velocities, and wave heights, the results provide valuable insights into how kelp motion varies in different environmental conditions.

2. Methods

2.1. Field Experiment

Two mature kelp individual plants, referred to as Plant 1 and Plant 2, with similar maturity (approximately 40 stipes), lengths, and distances from other kelp were chosen on the southern boundary of the McAbee kelp forest in Monterey, California, USA (36.6157°N, 121.8995°W, Figure 1a). McAbee is located on the eastern side of the Monterey peninsula and relatively well-protected from winds and waves. Starting at the holdfast, instruments were securely attached using zip-ties to each kelp individual at ~3.5 m intervals using a pre-measured line (Figure 1b). A small weight was attached to one end of the line to ensure taut length. A total of eight Ruskin RBRsolo³ pressure sensors (sampling rate of 2 Hz) and Loggerhead OpenTag3 MARGs were attached onto the two kelp individuals (see Table 1).

Loggerhead OpenTag3 instruments include pressure sensors (MS5837 or Keller PA7LD, sampling rate of 1 Hz) and IMUs (MPU9250, sampling rate of 20 Hz). Throughout this paper, we refer to the OpenTag3 IMUs as MARGs, which stands for magnetic, angular rate, and gravity. The MARGs were custom-made, with rechargeable lithium batteries encased in epoxy (approximate run time of 20 days). One RBR solo³ pressure sensor was deployed on the seafloor next to Plant 1 to measure depth and wave conditions. A Nortek Signature 1000 Acoustic Doppler Profiler (ADCP) sampling at 1 Hz was deployed in a sandy bed without vegetation approximately 20 m southeast of the kelp forest to avoid measurement contamination from any kelp fronds in proximity.

In this study, the baseline data from both plants were analyzed to understand how kelp tilt changes in different currents, waves, and water levels. The control individual (Plant 1) was unmodified during the duration of the experiment, which took place from 21 November 2022 to 8 December 2022. The surface canopy was removed from the impact kelp plant as an intervention action to understand how the presence of a surface canopy influences kelp motion. On November 30, the surface canopy stipes were removed from Plant 2, so that the entire plant length was 12 m. The approximate length of the removed surface canopy was 5 m. Instruments were recovered 8 days later on December 8; the entire deployment lasted 18 days. Two instruments failed during the experiment on Plant 2: the RBRsolo³ located at 0.25 m above bottom (mab), and the MARG located at 6.6 mab.

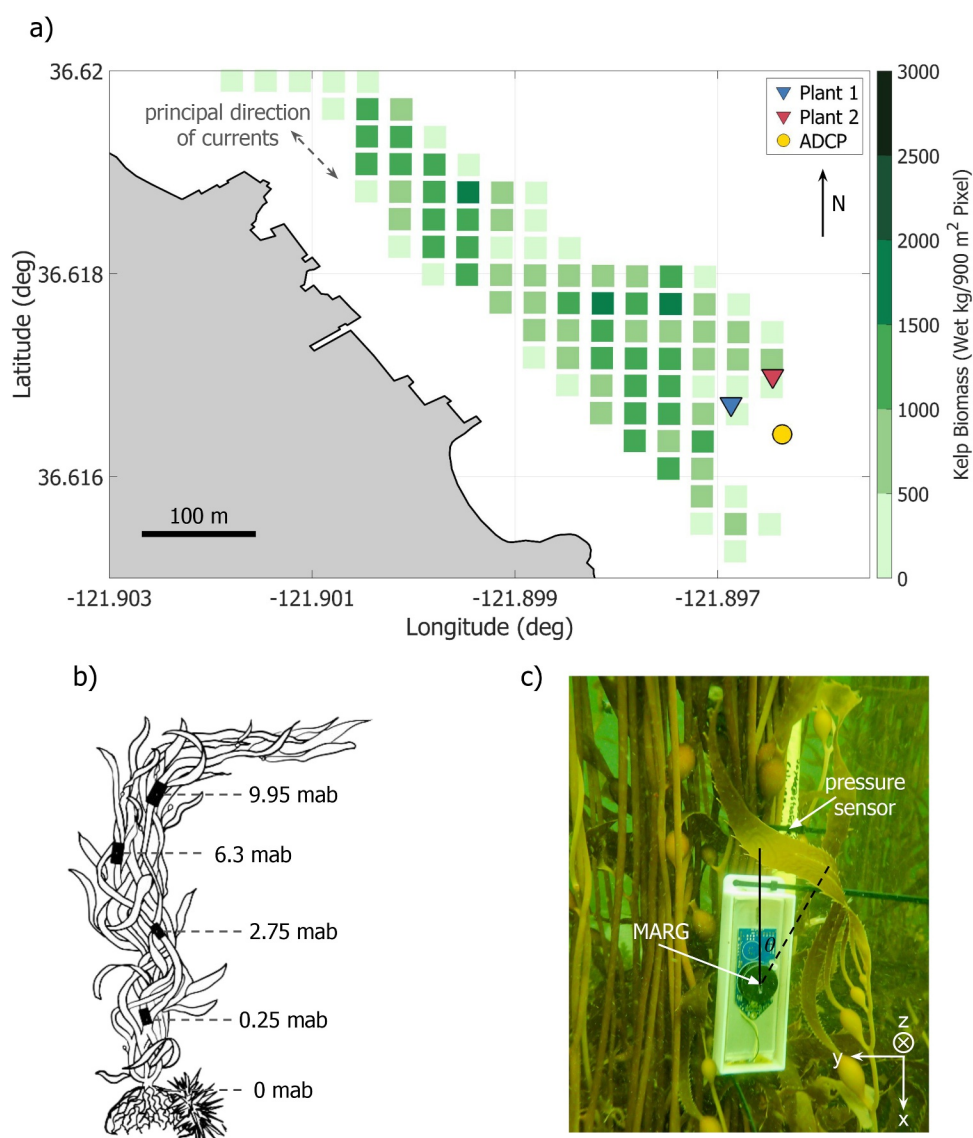


Figure 1. (a) Map of McAbee beach and instrument locations, where the shoreline is shown in gray. Green squares represent the average satellite-derived estimates of kelp biomass (wet kg/900 m² pixel) in 2022 (T. W. Bell et al., 2024). The blue and red triangles represent the locations of the control (Plant 1) and impact (Plant 2) sites, respectively. The ADCP location is denoted by the yellow circle. (b) Schematic showing the heights of instruments that were attached to Plant 1 (not to scale). Refer to Table 1 for the instrument locations on Plant 2. (c) Photo showing how the MARGs and pressure sensors are deployed with each other and the coordinate system of the MARGs.

After the experiment, the approximate heights above bottom of the sensors attached to the kelp individuals (Table 1) were calculated using data from the RBRsolo³ pressure sensors. Pressure measurements were low-pass filtered (frequency cutoff of 0.033 Hz) and compared with data measured at the seabed to find the maximum height above bottom. Assuming that there is no stretching in the kelp stipes, this maximum distance is then equal to the sensor's distance along the individual plant when the plant is near-vertical (no deflection). Near surface ADCP bins that had incomplete data due to depth fluctuations from tides were excluded from further analysis. Significant wave heights were estimated every hour as four times the standard deviation of the detrended sea-surface signal. Linear wave theory was used to correct for the frequency-dependent attenuation of the pressure signal due to depth, as our sensor was located at the seafloor. A pressure-correction factor with a cut-off frequency of 0.3 Hz was applied (Dean & Dalrymple, 1991).

Table 1

Heights Above Bottom of Instruments on the Control (Plant 1) and Impact (Plant 2) Plants, in Units of m Above Bottom (mab)

Plant 1 (m)	Plant 2 (m)
0.25	0.25
2.75	3.05
6.30	6.60
9.95	9.65

Note. Estimates of heights above bottom were obtained using data from the RBRsolo³ pressure sensors. The pressure sensor at 0.25 mab on Plant 2 and the MARG at 6.6 mab on Plant 2 both failed.

2.2. MARG Processing

Clock drift caused by temperature fluctuations was evident in the MARG data, but it was neither consistent nor predictable across different devices. Therefore, we used the pressure record from the RBRsolo³s that were deployed with each respective MARG to align the measurements within 1 s of the appropriate time. The clocks on the RBRsolo³ pressure sensors were accurate and had minimal discrepancies between sensors at the end of the deployment. As an additional check for time synchronization, divers also tugged on both kelp individuals when the surface canopy on Plant 2 was removed.

The Madgwick filter (publicly available and implemented in MATLAB®) was used to estimate orientation by combining data from the accelerometers,

gyroscopes, and magnetometers (Madgwick et al., 2011). The process of combining measurements from multiple sensors produces a more accurate estimate of the orientation than can be measured from any individual sensor alone. Here, the gyroscopes provide high-frequency measurements of angular rate that can be integrated to track orientation changes, but accumulate drift over time. The accelerometers and magnetometers provide absolute references of orientation relative to gravity and Earth's magnetic field, respectively, and are used to correct for gyroscope drift. The Madgwick algorithm uses a gradient-descent approach to minimize the discrepancy between the orientation predicted from angular rate integration and the orientation inferred from the accelerometer and magnetometer measurements. This compensates for integration drift and the weaknesses of each sensor (Madgwick et al., 2011). The output is a quaternion, a four-element mathematical object that compactly and uniquely represents three-dimensional orientation for each measurement in time.

The quaternion output describes the orientation of the sensor frame (x, y, z) relative to the global reference frame (north, east, down); see Figure 1c for an illustration of how the sensor frame is oriented during deployment. The ZYX Euler angle representation of the sensor orientation can then be obtained from the quaternion representation. The ZYX Euler angles-yaw (ϕ), pitch (θ), and roll (ψ)- describe the orientation of the sensor frame achieved by three successive rotations, starting from alignment with the global reference frame: first, by ψ about the z -axis, then by θ about the y -axis, and finally by ϕ about the x -axis. It is important to note that the Euler angle representation is non-commutative, and that θ represents a rotation to the sensor frame after it has already been rotated by ψ about the down direction (z -axis). Likewise, ϕ represents a rotation to the sensor frame after it has already been sequentially rotated out of alignment with the global reference frame by ψ and θ .

In practice, the yaw, which represents the rotation of the sensor frame about the down-axis, can be interpreted as the heading of the sensor, while the pitch can be interpreted as the absolute tilt angle from the vertical. The pitch tilts the sensor frame up and down relative to the heading. In the following sections, we present results on pitch (θ), where 0° corresponds with being vertically upright and 90° with the sensor oriented parallel to the ground. The range of θ is restricted to be between 0° and 180°. We were unable to produce extensive analysis on the other two Euler angles, yaw and roll, because the data suffered from erratic behavior due to Gimbal lock. Gimbal lock is a problem where the degrees of freedom in three dimensional space are reduced to two dimensions under certain rotations, such as when the MARGs are deployed upright and aligned closely with the vertical ($\theta \approx 0$ or 180°). Gimbal lock is not an issue for quaternions or rotation matrix representations (Hemingway & O'Reilly, 2018); the problem arises because Euler angle representations are not unique, and the same quaternion can map to different Euler angle combinations. As a result, we were unable to fully characterize the direction of motion in the horizontal plane using the Euler angles alone. We were only able to analyze the tilt angle from the vertical, independent of heading. This limitation is further explored in Section 4.

We also approximated tilt angle using data from the RBRsolo³ pressure sensors. Briefly, pressure measurements were low-pass filtered (frequency cutoff of 0.033 Hz) to produce the average difference between the distance above the bottom and the maximum distance above bottom from when the kelp plant is vertical. Assuming a simplified two-dimensional plane and that there is no bending or stretching in the plant, it is then possible to use that difference to estimate the average deflection from the vertical (see Supporting Information S1 for further explanation and a schematic), which is what the pitch represents. The average pitch measurements from Plant 1 were in high agreement with tilts estimated from the RBRsolo³ pressure sensors (shown as Figure S1 in

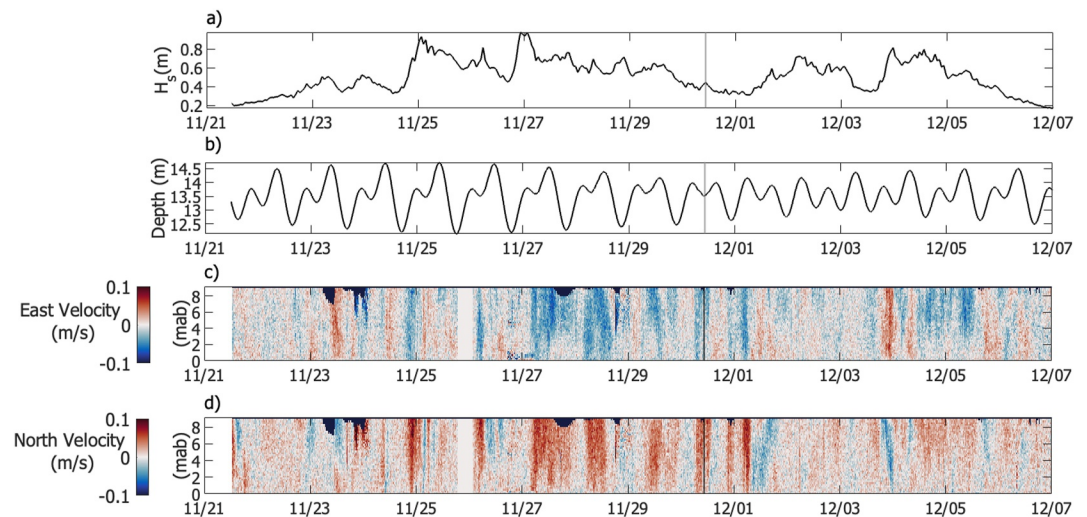


Figure 2. Hydrodynamic conditions during experiment: (a) significant wave height (H_s), (b) depth, (c) velocity in the east direction, (d) velocity in the north direction. Velocities shown are 15 min low-pass filtered. In all panels, the vertical line represents the start of the impact period, or when the surface canopy on Plant 2 was removed.

Supporting Information S1), indicating that the pitch data captures a majority of the variability in motion. We also used the tilt estimates from the closest working RBRsolo³ pressure sensor to correct a persistent offset (+14.7°) in the pitch measurement from the MARG located at 0.25 mab on Plant 2. The offset was evident in the entire timeseries and clearly unphysical; it is likely from a slight deflection in how the MARG was initially attached to the plant. Because the offset subtracted was a constant bias, it should not affect the following results on low-pass and band-pass filtered data.

We present analysis of the MARG pitch on two separate timescales using a low-pass filter and a high frequency band-pass filter to separate the influence of currents and tides from surface gravity waves. The Fourier filter was used for all filtering applications. We define our low frequency band to have a high frequency cutoff of 0.0033 Hz, and the high frequency band to have a low frequency cutoff of 0.033 Hz and high frequency cutoff of 2 Hz. The low frequency band captures signals with timescales longer than 5 min, whereas the high frequency band isolates signals occurring on timescales comparable to surface gravity waves, ranging from 0.5 to 30 s. Henceforth, the time series of low frequency tilts is referred to as θ_{lp} , and its hourly average as $\bar{\theta}_{lp}$. Similarly, the time series of high frequency tilts is referred to as θ_{hp} , and its hourly root-mean-square (RMS) as $\tilde{\theta}_{hp}$. $\tilde{\theta}_{hp}$ can be interpreted as being proportional to the average amplitude of tilting due to wave motion. We note that we were only able to characterize giant kelp motion using pitch data from the MARGs, which does not necessarily reflect distance traveled or velocity. Limited interpretations of this tilt data is addressed in Section 4.

3. Results

As described in the methods, one MARG out of four attached to the impact plant (Plant 2) did not collect data (located at 6.6 mab). Nonetheless, both plants had similar responses to tides, currents, and waves during the baseline period (for more information, please refer to Figure S3 in Supporting Information S1). The low-pass data (high frequency cutoff of 0.0033 Hz) was used to identify the kelp tilt response to forcings with longer timescales, such as tides, currents, and waves. The high-frequency data (low frequency cutoff of 0.033 Hz) was used to examine the tilt response to higher-frequency forcing like surface gravity waves. We then compare the motion of the impact plant during the control period and impact period to examine how the surface canopy impacts kelp motion.

3.1. Environmental Forcing Conditions

The maximum tidal range during the deployment was 2.6 m (Figure 2b). Depth-averaged currents were small, ranging from ± 0.1 m/s and visible with a 15 min low-pass filter (Figures 2c and 2d). Tidal and wind-driven alongshore currents dominated flow variability, with the strongest flow near the surface. Swell waves

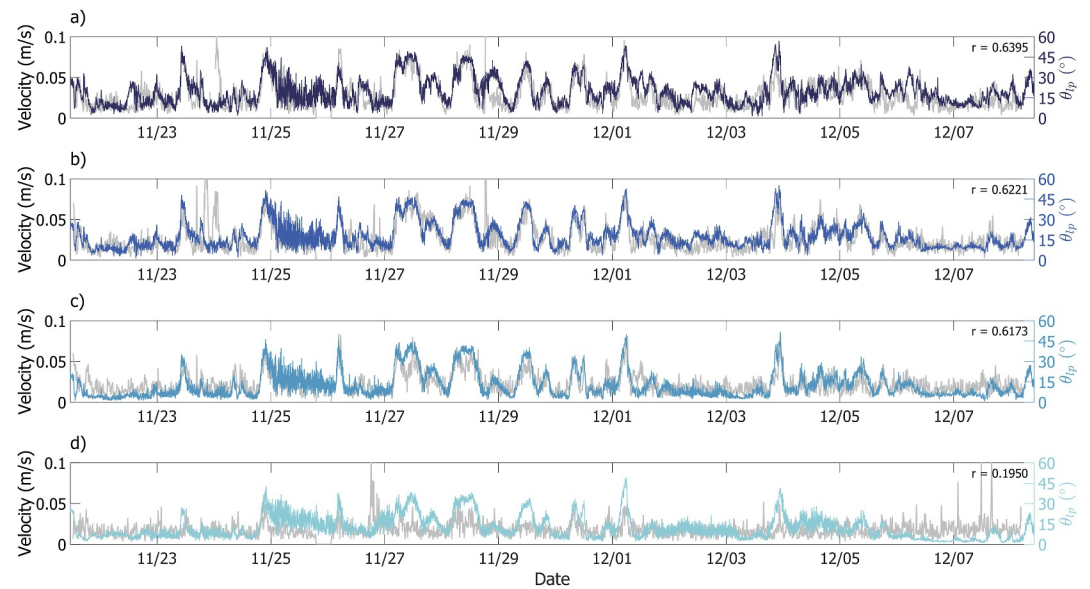


Figure 3. Low-pass filtered kelp tilts of Plant 1 (blue) and the 10 min averaged current speed (gray) at sensor depths of (a) 9.95 mab, (b) 6.3 mab, (c) 2.75 mab, (d) 0.25 mab. The frequency cutoff for the low-pass filter on the tilt data was 0.0033 Hz. Correlation coefficients between kelp tilt and current speed at each sensor depth are reported in the upper right.

dominated the high frequency wave forcing, with peak periods ranging from 7.1 to 16.7 s over the course of the experiment. Significant wave heights (H_s) varied over the experiment, ranging from calm conditions ($H_s < 0.5$ m) to a maximum of 0.98 m (Figure 2a). Currents, waves, and tidal fluctuations were slightly stronger in the first 10-day control period compared to the latter 9-day impact period. Mean depth-averaged current speed was 0.020 ± 0.016 m/s during the control period and 0.016 ± 0.011 m/s during the impact period. Mean H_s during the control period was 0.53 ± 0.18 m and 0.43 ± 0.18 m during the impact periods. The difference between maximum and minimum depth was larger during the control period (2.6 m compared to 2.1 m).

3.2. Response of Kelp to Hydrodynamic Forcing

Here, we present analysis on the kelp tilt response to hydrodynamic forcing of the control plant, Plant 1, since all MARGS were operational and results are similar to Plant 2. Low-pass filtered motions of the control plant reveal that it experienced mean tilts of up to 57° from vertical. These low-pass tilts are strongly correlated with the 10-min averaged current speed, here shown using currents binned at ± 0.1 m nearest to the average depth of the sensor (Figure 3). The magnitude of tilting correlates well ($r = 0.6395$ at the top sensor) with small changes (order of 0.01 m/s) in the velocities at the local depth. The maximum tilt recorded at the top sensor is 57° , compared to 49° (difference of 8°) at the bottom sensor. However, the bottom MARG (Figure 3d) recorded relatively strong tilts despite smaller current velocities at the local depth. For example, between November 27 and 29, the correlation between the tilt response and local current appears to be weaker at the bottom compared to the other sensors. Around December 1, maximum tilt was around 45° for sensors at every depth, but the current near the bottom sensor was only around 0.05 m/s compared to 0.08 m/s near the top. This mismatch between tilt and local current speed explains the overall weaker linear relationship at the bottom-most sensor ($r = 0.1950$) compared to sensors at other depths.

The RMS amplitude of the high frequency tilt of the control plant strongly correlates with the hourly significant wave height, particularly at the bottom most sensor (Figure 4, correlation coefficients are reported in Figure 5). The largest response to surface gravity wave forcing was near the holdfast, with hourly RMS tilts ($\tilde{\theta}_{hp}$) of up to 10° in relatively high wave conditions ($H_s = 1$ m). In comparison, the MARG sensor closest to the surface had a maximum $\tilde{\theta}_{hp}$ of around 4° , indicating that the near-bottom response of kelp tilt to wave forcing is around double the amplitude of tilting elsewhere along the plant. The responses of $\tilde{\theta}_{hp}$ from the other three MARGs in the water

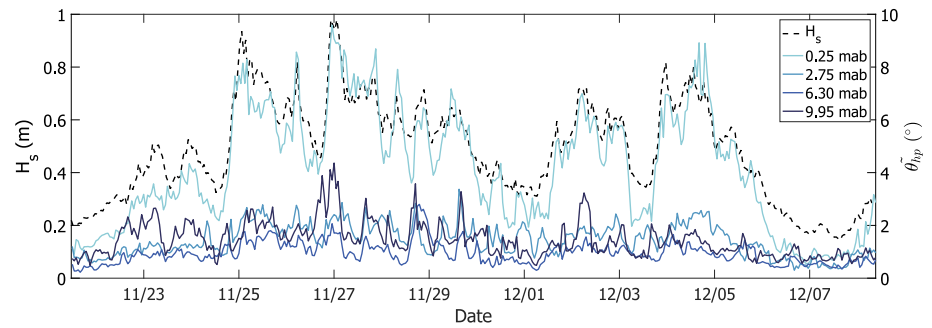


Figure 4. Hourly root mean squared of the high frequency kelp tilts of Plant 1 (blue) and significant wave height (black, dashed). Sensor height above bottom is denoted by the shade of blue. Low and high frequency cutoffs for the kelp tilt data were 0.033 and 2 Hz, respectively.

column (above the holdfast) were very similar overall, with marginally increased RMS tilts in the top sensor. The linear best fit for the bottom MARG is given by $\tilde{\theta}_{hp} = 11.31H_s - 1.4$, with $R^2 = 0.92$. Note that this best fit is only valid for $H_s \geq 0.13$, as outside of this range the equation leads to negative values of $\tilde{\theta}_{hp}$. The linear best fits

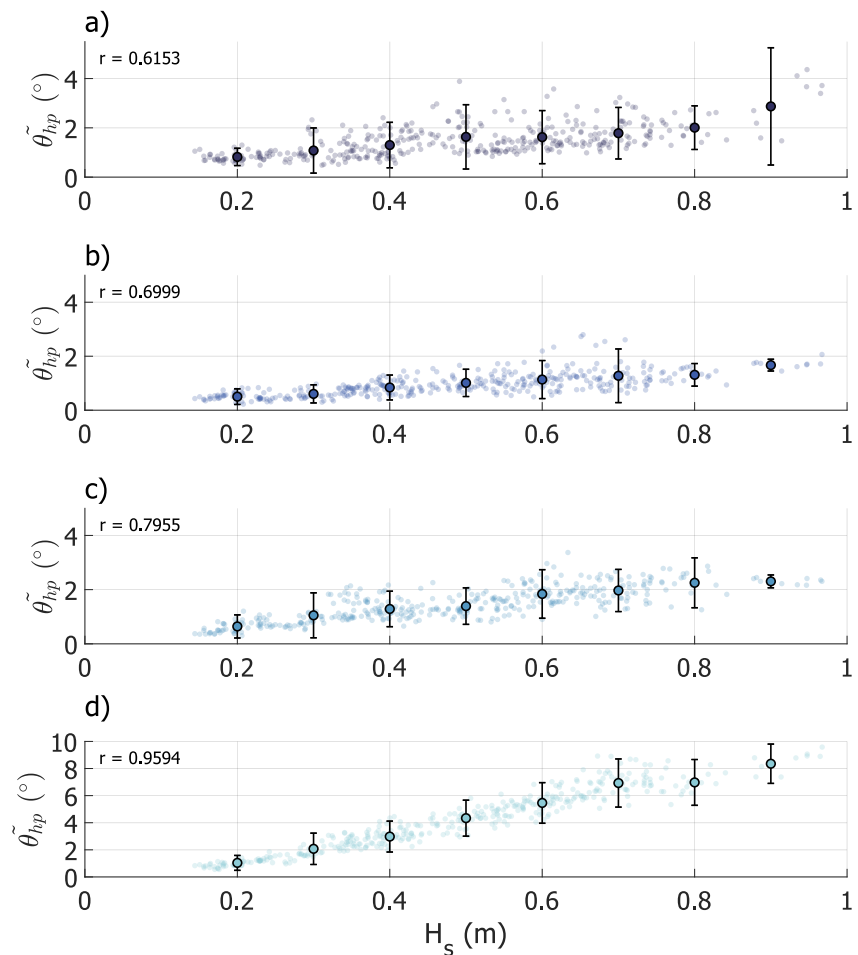


Figure 5. Hourly root mean squared (RMS) of high frequency kelp tilts vs. significant wave height at sensor depths of (a) 9.95 mab, (b) 6.3 mab, (c) 2.75 mab, (d) 0.25 mab. Solid points are binned data in 0.1 m intervals, with error bars representing two standard deviations. Note the difference in the y-axis scale in panel (d). The linear best fit for the MARG located at 0.25 mab (panel d, not shown) is given by $\tilde{\theta}_{hp} = 11.31H_s - 1.4$ ($R^2 = 0.92$, $H_s \geq 0.13$). Correlation coefficients between RMS high frequency tilt and significant wave height are reported in the upper left.

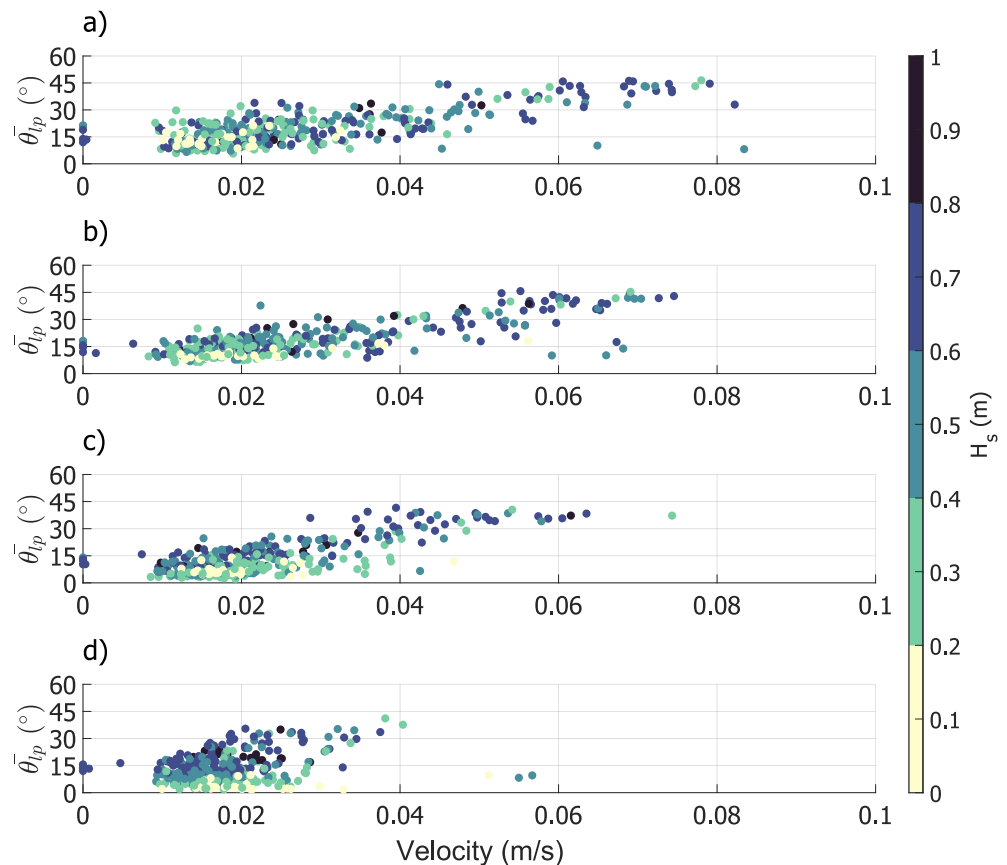


Figure 6. Hourly averaged low-pass filtered kelp tilts of Plant 1 plotted against depth-binned current speed. Sensor depths of (a) 9.95 mab, (b) 6.3 mab, (c) 2.75 mab, (d) 0.25 mab. Scatter points are colored by the significant wave height.

of the other three sensors are included in Table S1 in Supporting Information S1, but their R^2 values were substantially lower ($R^2 = 0.63, 0.49$, and 0.38 for 2.75, 6.30, and 9.95 mab, respectively).

Figure 5 demonstrates how increasing wave height affects the high frequency RMS pitch response. For small significant wave heights such as $H_s = 0.2$ m, the tilt response is minimal and uniform throughout the water column. However, for larger significant wave heights, we observe increasing tilts as a function of depth. At $H_s = 0.9$ m, we observe the greatest tilt deviations with depth, where the tilt response near the bottom was 4 times larger than the response at other locations in the water column. The strength of the linear correlation was not improved by using a combination of wave height and peak wave period as the velocity scale (not shown).

While it is valuable to look at waves and currents independently, these forcings are concurrent, and the relative contributions to creating tilt may vary in time. The mean position of the kelp depends not just on current strength, but also on wave forcing (Gaylord et al., 2003). Increasing wave heights can induce greater average tilts, especially near the bottom where the kelp response to wave orbital motion is the strongest and response to currents is the weakest (Figure 6). This effect is most visible in Figure 6d. When significant wave heights are small (e.g., $H_s < 0.2$ m), the near-bottom low-pass tilt is less than 10° . When significant wave heights are larger (e.g., $H_s > 0.6$ m), the near bottom low-pass tilt increases to around 20° , approximately twice as large as its value during weak wave conditions. The strength of the impact of wave height on low-pass tilts decreases with increasing distance from the bottom, consistent with the response of kelp motion to waves shown in Figure 4. This relationship between low-pass tilt and H_s likely contributes to the significantly weaker correlation between current speed and low-pass tilt at the bottom sensor.

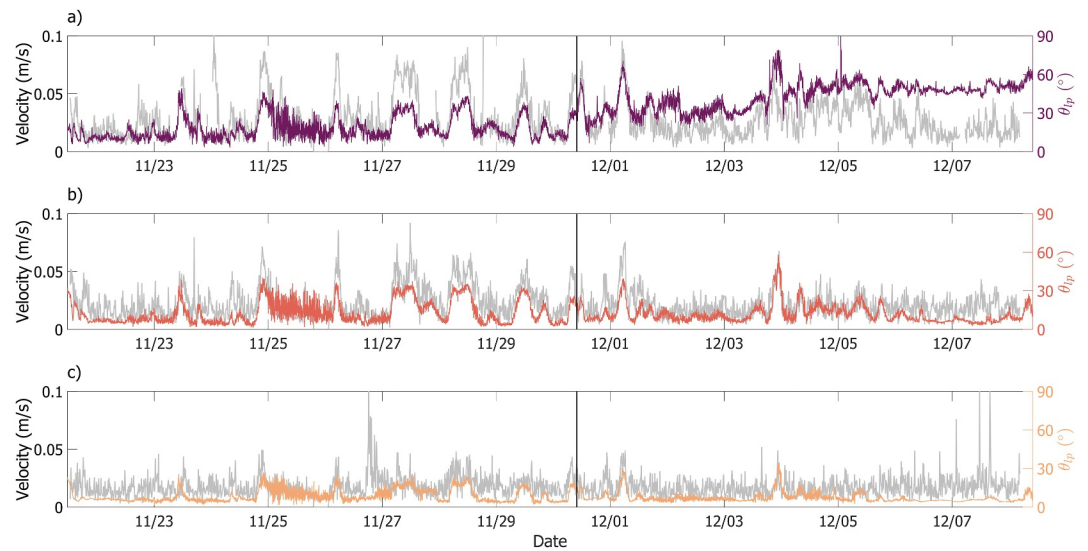


Figure 7. Low-pass filtered kelp tilts of Plant 2 (red) and 10-min averaged current speed (gray) around sensor depths of (a) 9.65 mab, (b) 3.05 mab, (c) 0.25 mab. Current speed in each panel is computed from depth bins within ± 0.1 m of each sensor's mean depth. The black line in each panel represents when the surface canopy was removed.

3.3. Impact of Surface Canopy Removal

During the second period of the experiment, the surface canopy of the impact plant (Plant 2) was removed (the intervention action). Figure 7 shows the time series of the low-pass tilt response of the impact plant with the currents during both periods of the experiment. As a point of comparison, the low-pass tilt responses of Plant 1 and Plant 2 over the entire deployment are plotted together in Figure S3 in Supporting Information S1. When the surface canopy was removed (November 30, denoted by the gray vertical line), there was a significant change in the response to currents in the top sensor (Figure 7a, 9.65 mab). Pre-intervention, low-pass tilts generally increased or decreased with currents, comparable to the control plant. Post-intervention, the overall tilt deviated from the observed trend, with larger tilts during periods of both high and low current speeds. Moreover, the maximum tilt observed (90.2°) was close to 90° , indicating that the top sensor was horizontal near the surface. This extreme value was from an isolated data point on December 5, but the average tilt of the top sensor over the second half of the experiment was still $43 \pm 10.8^\circ$. This value is substantially larger compared to a mean tilt of $17 \pm 8.6^\circ$ during the control period, reflecting the consistent trend of increased near-surface mean tilt after the surface canopy was removed. There does appear to be some drift in the data (increasing mean tilt with time), and we attribute this to a change in the mean position of the top sensor, rather than instrument malfunctions (refer to Supporting Information S1 for more discussion on drift in the data).

Similarly, after surface canopy removal, near-surface wave-induced motion was amplified on the intervention plant. Figure 8 shows the power spectra of tilt response from the top sensors of the two plants for swell wave frequencies. During the control period, the measured motion of the two plants were similar. In contrast, during the intervention period, increased tilting with swell waves was observed on the intervention plant. Increased tilt response was most pronounced for higher frequencies (0.05–0.2 Hz). At the peak wave frequency of 0.08 Hz, power spectral density of wave-associated tilt near the top of the intervention plant was significantly larger compared to the control plant, $90^\circ^2/\text{Hz}$ versus $23^\circ^2/\text{Hz}$, respectively. Wave conditions between the two periods were similar to one another (mean H_s was 0.53 ± 0.18 m during the control period and 0.43 ± 0.18 m during the impact period), implying that the change in tilt response on the intervention plant was due to the removal of the surface canopy.

As seen in Figure 9, during the control period, the hourly RMS of high frequency tilt response on the top sensor (9.65 mab) increased with significant wave height. The linear best fit during the control period is given by $\tilde{\theta}_{hp} = 2.91H_s - 0.20$ ($R^2 = 0.465$). During the intervention period, the high frequency tilt response was larger for all wave conditions, including small wave heights, resulting in a linear best fit of $\tilde{\theta}_{hp} = 2.64H_s + 1.09$.

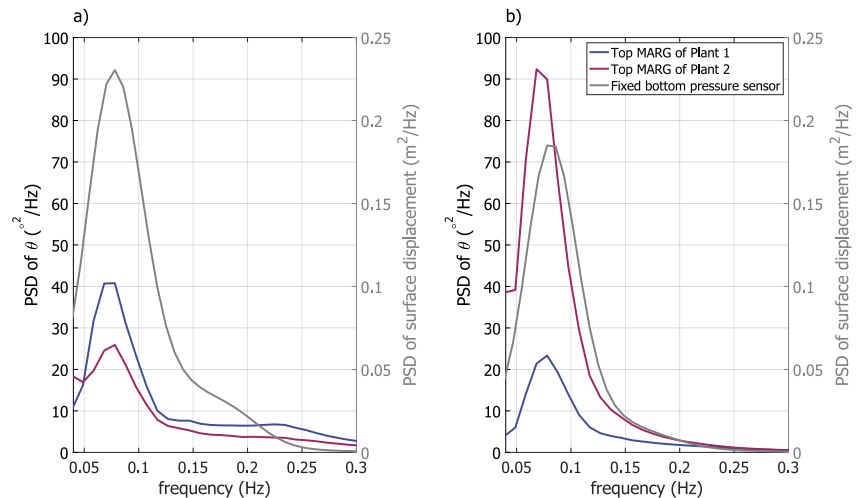


Figure 8. (a) Power spectral density of tilts from the top sensors of both Plant 1 (blue) and Plant 2 (purple) during the control period. (b) Power spectral density of tilts from the top sensors of Plant 1 and Plant 2 after the surface canopy was removed (impact period). In both panels, the power spectral density of the swell wave surface displacement (gray) measured by the bottom pressure sensor is included for reference. Power spectral density was calculated using Welch's method, where the signal was divided into Hamming-windowed data segments with 75% overlap.

($R^2 = 0.445$). 95% confidence intervals for the slopes and intercepts of the linear best fit models can be found in Table S2 in Supporting Information S1. The intercepts of the linear best fits of the two periods significantly deviate, indicating that without the surface canopy, the top of the plant was free to move more flexibly with the waves. However, similar slopes suggest that tilting increases with wave height at relatively similar rates before and after removal. We expected that the standard deviation of $\tilde{\theta}_{hp}$ would approach zero with decreasing H_s . However, the standard deviation within the smaller wave-height bins is notably higher during the impact period, particularly seen in the 0.3 and 0.4 m H_s bins. In the first 1–3 days after the surface canopy was removed, both the mean tilt ($\tilde{\theta}_p$) and $\tilde{\theta}_{hp}$ had a similar range of values to what was measured during the control period. Over the course of the impact period, as $\tilde{\theta}_p$ increases, so does $\tilde{\theta}_{hp}$ (shown by the color of the points in Figure 9). This explains why the standard deviation does not approach zero at small wave heights: during the impact period, the high frequency tilt measurements exhibited higher variability because they were increasing with time regardless of the wave regime. For comparison, the same data during the two periods from the top sensor of Plant 1 is presented as Figure S5 in Supporting Information S1.

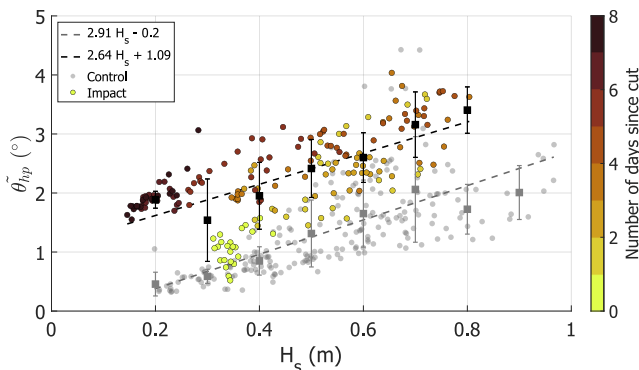


Figure 9. Hourly root mean squared of high frequency kelp tilts from the MARG sensor closest to the surface on Plant 2, as a function of significant wave height. Gray points are from before the surface canopy was cut, colored points are from after. The shade of the post-intervention (colored) points represents the number of days since the cut. Square points are binned data in 0.1 m intervals, with error bars representing ± 1 standard deviation. Dashed gray ($R^2 = 0.465$) and black lines ($R^2 = 0.445$) represent the linear best fits for before and after intervention, respectively.

In contrast to the top sensor on the impact plant, the bottom two sensors exhibited little to no change in the tilt response to currents or waves. Because the sensor on Plant 2 located at 6.6 mab failed, it remains unclear if the removal of the surface canopy impacted mean tilt or tilting with the waves at greater depths. Overall, the observations indicate that without the surface canopy, kelp motion with the currents and waves increases near the surface, while kelp motion near the bottom is not impacted.

4. Discussion

We observed kelp motion over multiple timescales and for a variety of hydrodynamic conditions. In general, tilt response in currents is predominantly pronation with minimal bending, as low-pass tilts were relatively uniform along the length of the control plant. However, velocities were stronger closer to the surface, resulting in larger low-pass tilt responses near the surface canopy compared to the holdfast, accounting for a maximum of 15% difference in tilt angle between the top and bottom MARG sensors. Therefore,

some bending was observed, but it was small relative to the magnitude of pronation along the individual. These tilt response observations are similar to what was observed in Mullarney and Pilditch (2017) for infragravity wave frequencies. Increased tilt near the seafloor during periods of relatively high currents (e.g., December 1) suggest that the tilt response at the bottom is affected not only by the localized current, but also by kelp motion dynamics in the water column. The drag forces from higher up the kelp force the entire kelp to tilt. Over longer timescales, the blades near the bottom of the plant are also more free to fully stretch out. Blade elongation minimizes stipe bending rather than minimizing tilt, which may be an evolutionary strategy to help reduce internal stresses from hydrodynamic forces (Utter & Denny, 1996). We also found that the motion at each MARG height had a stronger correlation with the closest binned velocity (e.g., at the sensor's average depth) compared to the depth-averaged velocity (shown as Figure S4 in Supporting Information S1), indicating that any shear in the velocity might have effects on kelp motion as well.

Like Mullarney and Pilditch (2017), we observed that the largest tilt response to swell waves occurs near the bottom of the plant. There was a slight increase in tilting with the waves recorded by the sensor closest to the surface relative to other mid-water column sensors; this may be due to enhanced wave orbital motion at the surface. In contrast to the pronation response in currents, we observe bending in the kelp as it moves with the wave orbitals, which is evidenced by the significantly larger tilt response near the bottom compared to elsewhere in the water column. The overall behavior of kelp motion in response to forcing by surface gravity waves is consistent with what is theoretically predicted by Henderson (2019), who found that bending for highly buoyant and flexible vegetation like giant kelp is concentrated in the near bed region. Above the near-bed boundary layer, plants move passively with the surrounding water, such that drag production is minimal, which implies that the wave motion close to the holdfast is important for understanding the contribution of giant kelp to enhancing bottom stress.

Removing the surface canopy leads to a significant loss of buoyancy for the entire kelp plant, resulting in increased flexibility and more motion in response to mean currents and waves. Pneumatocysts are present at the base of every blade along the fronds of each kelp individual. The dense frond arrangement in the surface canopy provides an upward buoyancy force and tension along the stipes. When the surface canopy is removed, the near-surface section of the plant gains additional degrees of freedom when there are no longer any tensile forces from the surface canopy to restrict motion. Over time, these effects compound and result in near-surface plant extension. In contrast with the control period, there is now bending in the water column rather than rigid pronation. Without the surface canopy, the top of the kelp individual is more sensitive to any background wave action and experiences greater amplitude oscillations. Additionally, given the high standard deviations and change with time of θ_{tp} after the surface canopy removal, the mean tilt angle of the plant may influence the amplitude of wave-induced oscillations. We hypothesize that a kelp plant with an increased mean tilt may also have reduced restoring forces from buoyancy at the surface, which allows for greater wave-induced oscillatory motion. However, the data collected in this experiment was limited and further research is required to draw more concrete conclusions about these dynamics. The post-intervention motion observed is consistent with the simulations of individual fronds with no surface canopy from Meng and Chamecki (2025), where wave-induced tilting is strongest at the bottom and at the free end of the kelp. They also observed significant bending and more complex modes of motion in their model kelp.

At our site, wave orbital velocities were more pronounced than current velocities. The mean depth-averaged wave orbital velocity over the entire experiment was 0.13 ± 0.05 m/s, while the mean magnitude of depth-averaged low-pass filtered velocity was only 0.018 ± 0.014 m/s. The two velocity scales generally differ by around an order of magnitude, although there were periods where the mean depth-averaged current was up to 0.1 m/s. The strongest tilt response occurred in “high current” conditions. The difference in surface gravity wave and tidal current timescales could account for the observed effect, where the longer timescale of currents allows kelp to tilt over time. In periods of increased surface gravity wave activity, the enhancement of tilts in response to currents is similar to other wave phenomena, such as canopy-forming kelp extending out more fully due to the increased transport associated with Stokes Drift (Gaylord et al., 2003), or asymmetric motion in flexible aquatic vegetation (Zhu et al., 2020). Stronger wave heights for a given wave period generally have larger wave orbital excursion; in these conditions, the blades near the bottom are allowed to reach full extension. This exerts a stronger net force and leads to additional mean tilt (Mullarney & Pilditch, 2017). While currents generally induce the greatest degree of motion, when the surface canopy is removed, waves have a stronger impact on kelp motion.

For our measurements, we were unable to discern effects from tidal depth on kelp motion in the water column. Given the tidal range, the length of the surface canopy would have changed by approximately 2 m from low to high tide. We hypothesize that the minimal effects of tidal depth variations was because the surface canopy on both the control and impact plants had already reached critical mass, for example, the buoyancy was sufficiently large even at high tide, such that changes with the tidal range had minimal impacts on kelp motion in the water column. Likewise, when the surface canopy was removed on the impact plant, tidal depth changes did not add sufficient dampening or buoyancy to affect motion in the water column. Note that the top-most sensors on the two plants were still deployed ~3 m below the lowest depth recorded, so it is possible that changes in tidal depth did affect the positioning of near-surface portions of the plant before the intervention action.

The results presented above allow us to form hypotheses about how the motion of giant kelp affects drag and wave dissipation. First, the surface canopy is an important control for the drag of giant kelp. Because of the increased length of the surface canopy, pronation of kelp with currents increases the amount of biomass in the water column and has no change on the frontal area of stipes exposed to the flow. Under the assumption of no bending, the fractional increase of biomass the water column with tilting is given by $1/\cos(\theta) - 1$ (see Supporting Information S1). For the maximum observed depth-averaged tilt of 51.3° on the control plant, this amounts to an additional 8.5 m of kelp plant in the water column compared to if it were vertically upright (note that the exact length also depends on the depth). Thus, reconfiguration of giant kelp with the mean current leads to the presence of more drag elements, that is, stipes and blades, in the water column. It was also posited by Meng and Chamecki (2025) that the blades of giant kelp only experience a minimal amount of streamlining under reconfiguration. For submerged seagrasses, reconfiguration with the currents typically results in drag reduction via streamlining and decreased frontal area (De Langre, 2008). We hypothesize that unlike seagrasses, the reconfiguration of giant kelp may only have minimal effects on frontal area and thus minimal drag reduction in the water column. Moreover, the fronds of an individual kelp plant are entangled and generally move together as a whole. The combined buoyancy and resistance to bending of a collection of fronds, that is, a single kelp plant, is greater than that of an individual frond. Thus, kelp models for a single frond (e.g., Meng & Chamecki, 2025; Strong-Wright & Taylor, 2024) may represent juvenile kelp individuals but overestimate a mature kelp individual's tilt response to currents and waves, especially near the surface. The drag of giant kelp is also affected by blade dynamics, which were not addressed in our study. Strong currents can induce the streamlining of blades parallel to the direction of the current, which reduces the frontal area of the blades (Koehl et al., 2008). Further study is required to understand the timescales of the blade response to currents and waves and the relative contribution of the blades to drag. In addition to increasing surface area, the presence of the canopy also limits relative motion with the surrounding flow in the water column. The inhibitive effect of the surface canopy means that theoretical and numerical models for the motion of flexible vegetation with no surface canopy (e.g., Henderson, 2019; Luhar & Nepf, 2016; Meng & Chamecki, 2025) may over-predict the amount of drag reduction with flexible motion for giant kelp. As discussed in Mullarney and Pilditch (2017), the impact of the surface canopy on motion and drag is strongest near the surface, further adding to the depth-dependent nature of energy dissipation by giant kelp.

Second, the motion of giant kelp with surface gravity waves may significantly impact the drag production of kelp forests. Our results indicate that there is considerable wave-induced motion throughout the entire water column. Somewhat counterintuitively, the high degree of tilting with waves near the bottom supports the conclusions of Henderson (2019) and Meng and Chamecki (2025) that drag is primarily generated in a bottom boundary layer (the bottom 50% of the plant). Similarly, Rosman et al. (2013) had proposed a model for the drag of giant kelp, where it was assumed that the surface canopy moves exactly with the waves and that there is no motion at the holdfast. It is important to note here that tilt measurements are not the same as relative motion with the surrounding flow; rather, they reflect local deflections from the vertical. Giant kelp forests likely add to the enhancement of bottom friction and drag on currents by wave-current interactions (Grant & Madsen, 1979). As a result, hydrodynamic models of flows through kelp forests that neglect the contribution of surface gravity waves to the drag forces of kelp may underestimate current attenuation (Rosman et al., 2013).

4.1. Study Limitations

A significant limitation of our study is that it is not possible to use the tilt measurements to deduce the distance moved or velocity of the kelp. However, the distance or relative velocity with the flow is the quantity that is used to parameterize for the drag on currents or dissipation of wave energy by kelp. The tilt measurements presented

above can only be interpreted as describing the spatial structure of the kelp plants' motion. While our measurements show that the amplitude of tilting with surface waves is weaker near the surface compared to the bed, this may be because a majority of the motion at the top is from translation, where the kelp is passively being moved with the surrounding flow without changing its tilt. Given that each stipe is like a continuous cable extending from holdfast to surface, any motion at the bottom should also affect the upper sections of the kelp; thus, the absolute distance traveled should generally be larger near the surface. Higher spatial resolution in the MARG measurements would more clearly resolve how tilt changes along the length of the kelp, which would enable better estimations of how deflections from the vertical correspond with horizontal motion. Future work on measuring instantaneous relative velocities and plant forces would also allow us to more explicitly link observed plant motion to drag production.

Because of gimbal lock due to the orientation of the MARGs during deployment, the pitch measurements presented here do not resolve the heading and motion in the horizontal plane. While the majority of the motion was captured in the pitch, we were unable to parse out the effects of any directional variability in the flow from the other Euler angles, for example, when currents and waves are in the same direction, opposite, or oblique. Gaylord et al. (2003) found that currents that are perpendicular to the direction of wave propagation could significantly mitigate the forces acting on canopy-forming kelp, so the relative directions of waves and currents could be important for the degree of tilting. Moreover, only absolute values for the pitch were reported (no negative angles). This did not present as a significant issue in our data analysis because the plants were deflected from the vertical for the majority of the experiment and the amplitude of tilting with the waves is relatively small, but it is possible that some aspects of symmetric motion about the vertical were unresolved. In the future, it may be possible to use the direct quaternion outputs, which are not affected by the issues of gimbal lock, to retrieve information regarding the directionality of kelp motion, but such analysis falls outside of the scope of this manuscript.

Our measurements were at a relatively low vertical resolution, with a sensor placed approximately every 3.5 m in 14.5 m of water. The near surface sensors were still well below the low tide depth, so we were unable to capture any possible effects due to tidal depth or resonance in plant motion (Meng & Chamecki, 2025; Mullarney & Henderson, 2010). As the experiment did not directly measure the motion of the surface canopy, we can only hypothesize how its presence affects drag generation and wave dissipation. The surface canopy, which contains a large quantity of a plant's total biomass, has more buoyancy and drag elements than in the water column. However, wave orbital velocities are strongest at the surface, and if the surface canopy is moving with the waves (e.g., relative velocities are low), then they may have little effect on current and wave attenuation. Additional measurements of giant kelp motion at the surface canopy would allow for improved hypotheses about how the presence of surface canopy affects the drag of giant kelp forests.

This study focused on measuring the motion of isolated kelp individuals at the edge of the McAbee kelp forest to better understand baseline dynamics without the added complications of more complex flows or plant-plant interactions, but flows within kelp forests are highly variable in space and time. Currents are attenuated within kelp beds but accelerated alongside the boundaries; dense patches of kelp can also promote flow variability via flow ducting through less dense areas (Gaylord et al., 2007; Valle-Levinson et al., 2022). Since plant motion is well-correlated with local hydrodynamic conditions, kelp at different locations within a forest may experience different patterns of motion. Our site was characterized by strong wave activity relative to currents; therefore, kelp in more current-dominated sites or during periods of reduced wave energy are likely to experience different dynamics. Moreover, the ADCP was deployed approximately 20 m outside the kelp forest, and the measured velocities may therefore overestimate the currents experienced by the instrumented kelp individuals. Drag from the surrounding kelp canopy is known to attenuate flow within kelp forests (e.g., Rosman et al., 2007), so the correlations between tilt and current speed reported here represent an upper bound on the true ambient flow driving kelp motion. Nonetheless, the observations of a strong response in tilt to relatively weak currents (<0.1 m/s) presented here imply that kelp at more exposed sites may have increased pronation. This could have consequences for turbulence generation, tensile stresses and breakage rates, and nutrient uptake, all of which depend on kelp motion (Denny et al., 1997; Huang et al., 2011; Hurd, 2000; Rosman et al., 2013; Stevens & Hurd, 1997; Utter & Denny, 1996). We also did not study tilting on timescales between 30 s and 5 min, as they include a constantly changing combination of influences from waves and currents and the differences are difficult to isolate. It could be of value to study these frequencies in the future, particularly for separating out the effects of infragravity versus surface gravity waves on kelp motion and drag.

Kelp motion may also be affected by entanglement with stipes from surrounding plants (Koehl & Wainwright, 1977). Clusters of kelp individuals are common in dense, perennial kelp forests, and it is likely that groups of individuals would have different responses to hydrodynamic forcing. Our experiment occurred during the northern hemisphere winter (December), when kelp senescence has largely already taken place (North, 1994). Both plant density and surface canopy length are at a minimum during this time, so the measured plants may have had more mobility in wavy flows and tilting with currents compared to plants in the summer, when biomass is at a maximum. Plants at different life stages, for example, younger adults, would also have fewer stipes or more flexibility than the two measured in this study. Future work will focus on studying the effects of these more complex site dynamics, plant interactions, and seasonal changes on kelp motion.

5. Conclusion

In this study, we conducted a field experiment to investigate the tilting and bending of giant kelp plants in response to varying current and wave conditions. Using IMUs, we observed significant kelp plant motion on multiple timescales. Plant position is most impacted by current speed, which causes extended periods of pronation or uniform tilt. Kelp also oscillates with the waves, with a depth-dependent response that is greatest near the seafloor. The surface canopy, one of the features unique about giant kelp, restricts motion induced by currents and the waves in the upper half of the water column. The surface canopy is critical for understanding how drag from kelp forests varies with growth stage, seasons, and tides, which affects current attenuation, horizontal transport, and vertical mixing in kelp forests (Rosman et al., 2010).

The results presented here have numerous implications for understanding how plant motion might affect flow and transport within kelp forests. In particular, the presence of a surface canopy and the response to surface gravity waves appear to be important for plant motion, and therefore, drag production and turbulence generation by kelp. Coastal ocean models that neglect plant motion, the presence of a canopy, or swell waves may not accurately capture the subtleties of flows through kelp forests. Further observations of kelp motion at different sites and kelp forest hydrodynamics in conjunction with improved modeling efforts are needed to better understand how *M. pyrifera* and other canopy forming macroalgae dissipate energy from the nearshore environment.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The data used in this study are available with the Stanford Digital Repository and can be accessed via <https://purl.stanford.edu/xn922bn0443> (Daly et al., 2025). The satellite-derived estimates of kelp biomass are available from the Environmental Data Initiative (T. Bell, 2023). Shoreline products are available from NOAA National Geodetic Survey (NOAA, n.d.).

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